



Faculty of Engineering, Built Environment and Information Technology

Fakulteit Ingenieurswese, Bou-omgewing en
Inligtingtegnologie / Lefapha la Boetšenere,
Tikologo ya Kago le Theknolotši ya Tshedimošo

Study Guide

Department of Electrical, Electronic and Computer Engineering

Electricity and Electronics EBN111

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TABLE OF CONTENTS

Contents

1. WELCOME	4
2. EDUCATIONAL APPROACH	4
2.1 General approach	5
2.2 Online approach	5
2.3 Hybrid approach	6
3. ARTICULATION WITH ECSA GRADUATE ATTRIBUTES AND OTHER MODULES IN THE DEGREE PROGRAM	6
4. LECTURERS, VENUES AND CONSULTING HOURS	6
4.1. Location of the Laboratories	6
4.2. Consulting hours	6
4.3. Timetable.....	7
5. STUDY MATERIALS AND PURCHASES	7
5.1. Prescribed textbooks	7
5.2. Components	7
6. LEARNING ACTIVITIES	8
6.1. Contact time and learning hours	8
6.2. Lectures	8
6.2.1 General approach to lectures.....	8
6.2.2 Face-to-face lectures	8
6.2.3 Online lectures	9
6.3. Hybrid teaching and learning approach	9
6.3.1 Hybrid teaching and learning	9
6.3.2 Online teaching and learning	9
6.4. Tutorial classes	9
6.4.1 General approach to tutorial lectures	9
6.4.2 Face-to-face tutorials	9
6.4.3 Online tutorials.....	9
6.5. AMS assignments.....	9
6.6. Laboratory work	10
6.6.1 General approach and practical guide.....	10
6.6.2 Approach to online practicals.....	11
7. RULES OF ASSESSMENT.....	11
7.1. Requirements, subminima and calculation of marks	11
7.2. Assessment details.....	13
8. GENERAL	13
9. MODULE OBJECTIVES AND LEARNING OUTCOMES.....	14
9.1. Content and purpose of the module.....	14
9.2. Module Outcomes	14
9.3. Cognitive level of assessment.....	15
10. MODULE STRUCTURE	15

11.	STUDY THEMES.....	17
11.1.	STUDY THEME 1: Basic Concepts	17
11.2.	STUDY THEME 2: Basic laws	17
11.3.	STUDY THEME 3: Methods of Analysis.....	18
11.4.	STUDY THEME 4: Circuit Theorems.....	19
11.5.	STUDY THEME 5: Operational amplifiers.	19
11.6.	STUDY THEME 6: Capacitors and Inductors.....	20
11.7.	STUDY THEME 7: Sinusoids and Phasors	20
11.8.	STUDY THEME 8: Sinusoidal Steady-State Analysis	21
11.9.	STUDY THEME 9: Practicals	21
12.	PRACTICALS.....	22

ORGANISATIONAL COMPONENT

This guide serves as Part 2 of the study guide for this module and provides content that is specific to the learning content of the module. Part 1 of the study guide is a General EECE Study Guide available from the EECE Undergraduate ClickUP page or at bit.ly/EECEGeneralUGGuide and provides rules and policies that are applicable to all undergraduate modules presented by the Department of Electrical, Electronic and Computer Engineering.

For EECE students: if you cannot access the current-year EECE Undergraduate Students ClickUP module, please arrange access as soon as possible by completing the form at bit.ly/EECEClickUPAccess. All undergraduate-related departmental policies, online resources and communication are managed through this platform.

For students outside the Department of EECE: you do not need access to the EECE Undergraduate Students ClickUP as links to relevant documents and online forms are provided in this study guide. Relevant communication via the EECE Undergraduate Students ClickUP will be reposted on this module's ClickUP Announcements page.

1. WELCOME

The general objective of this module is to develop expertise in analysing and design electric and electronic circuits. The module is primarily focused on solving new problems based on known theoretical techniques, as opposed to the memorising of recipe-type procedures, or solving problems by simply inserting values into formulae. A thorough knowledge of the fundamental theory of circuits will be required. Students will continuously be presented with problems that they might not have seen before, but for which they will have been equipped with the necessary theoretical background. Students are strongly advised to gain as much experience as possible in solving circuit problems from the beginning of the module.

2. EDUCATIONAL APPROACH

“For the sole true end of education is simply this: to teach [students] how to learn for themselves; and whatever instruction fails to do this is effort spent in vain.”

Dorothy Sayers

[The Lost Tools of Learning](#)

The tools of learning progress as follows: thinking, argumentation, and expression. In the first step a student is shown what the concept/theory/principle is, secondly why and how to use it, and finally the student is allowed to express their knowledge through application to a general problem. More emphasis is placed on *why*-questions rather than *how*-questions. The aim is not only for students to master a specific technology or technique (*how*), but also to become masters of principles and methods so that they may pioneer technologies and techniques (*why*).

A good engineer is one who not only mastered the art of learning, but also enjoys the process of learning new material. Therefore, the lecturers in this module regard their roles as teaching students

how to learn for themselves and how to creatively apply their newly acquired skills. If an engineer is to be successful in this world, they should be able to understand the intricacies of a problem, appreciate the underlying assumptions, apply new knowledge in a critical and creative manner, and solve the problem within the logical boundaries of scientific principles.

2.1 General approach

The module follows a lecture schedule that will be rigorously adhered to throughout the semester as released on ClickUP. The schedule indicates the following:

- the section of the textbook to be discussed during a lecture,
- the set of tutorial problems for each tutorial class,
- the class test and semester test dates,
- the deadlines for assignments, and
- the week in which practicals will be completed with corresponding pre-practical lectures.

Students are expected to prepare for a lecture by reading through the section of the textbook which will be covered during the lecture. Students should prepare basic notes in preparation for the lecture and attempt the Example Problems. The aim of a lecture is to illustrate and explain the difficult concepts in the textbook. Students can supplement their notes during the lecture. After a lecture, a student is expected to complete the Practice Problems to ensure they understand the concepts associated with a specific textbook section.

Students are expected to complete the tutorial problems in preparation for the tutorial lectures. Since the tutorial lectures will focus on the most difficult problems, not all the problems will be covered during the lectures. However, a complete memorandum will be available after the tutorial class. Students are encouraged to make queries on the Discussion Board on ClickUP for problems they struggle with.

Should students do not understand the work after attending lectures, working through Example and Practice Problems in the textbook, and working through the tutorial problems, they are welcome to make an appointment with the Lecturers or Assistant Lecturers for consultation. The procedure for consultation is communicated on ClickUP. Lectures will however not be repeated during consulting hours. Students must bring their own attempts at solving the problem to the appointment.

There is a set of assignments associated with each chapter in the textbook. These assignments will be made available online and should be completed in the student's own time before the submission deadline as shown in the lecture schedule. Students are encouraged to make queries on the Discussion Board on ClickUP for problems they struggle with.

Information regarding the practicals will be communicated on ClickUP.

2.2 Online approach

The same approach as detailed above will be followed in the case where lectures cannot be presented in-person and on campus except for the following:

- pre-recorded lectures will replace the in-person lecture sessions,
- pre-recorded tutorial videos will replace in-person tutorial sessions,
- online consultation will replace in-person consultation, and
- tests will be written according to a procedure that will be communicated via clickUP.

2.3 Hybrid approach

While all lectures, tutorials and practicals will be presented face-to-face in person and on campus, assignments will be completed online. Note that video recordings of lectures and tutorial classes will not be made available.

3. ARTICULATION WITH ECSA GRADUATE ATTRIBUTES AND OTHER MODULES IN THE DEGREE PROGRAM

The programs presented by the Department of Electrical, Electronic and Computer Engineering have been designed to meet the eleven ECSA graduate attributes (GAs) that qualify the competencies required by professional engineers. While most modules contain at least a small component of a number of ECSA GAs, only the main attributes that are addressed in a module are indicated for the specific module.

This module is presented at entry level for GAs 1 and 2.

Refer to the General EECE Study Guide (Part 1) for a complete overview of the association of ECSA GAs with the modules in each degree program.

4. LECTURERS, VENUES AND CONSULTING HOURS

	Name	Office	Contact details	Consultation
Lecturers	Prof. Xianming Ye	Eng 1:14-5	012 420 4353 xianming.ye@up.ac.za	Please make an appointment via e-mail
	Mr. Joshua Thistleton	Eng 3:7-33	Joshua.thistleton@up.ac.za	
	Mr. Leon Staphorst		leon.staphorst@gmail.com	
Assistant lecturers	Refer to module ClickUP page			
EECE undergraduate administration	Ms Bainca Johnson Ms Mari Ferreira	Eng 3:7-18 Eng 3:7-19	eece.undergraduate@up.ac.za	
Faculty Student Advisor		Your Faculty Student Advisor can advise you on goal-setting, adjustment to university life, time management, study methods, stress management and career exploration. Book an individual consultation or attend a workshop. For other support services see Paragraph 12 in the EECE General Study Guide.		

4.1. Location of the Laboratories

All practicals take place in the CAEC lab (Eng 2: 4-39) for on-campus practicals. Detailed practical attendance schedule are posted on ClickUP.

4.2. Consulting hours

Refer to the section on consultation and communication with lecturers in the General EECE Study Guide.

Kindly refer to the module ClickUP page for specific arrangements about consultation hours. Students may consult lecturers and assistant lecturers only during the consulting hours as indicated, or by appointment. Consultation can either be in-person or via a suitable online platform as the situation demands.

This policy also holds before tests and exams. In other words, lecturers and assistant lecturers are only available during their normal consulting hours on the day before a test or examination. This policy aims to encourage students to plan their work and to work continuously.

4.3. Timetable

The lecture timetable can be found on your student portal, the table below, or via the link to <https://minstro.up.ac.za/timetables/>.

Module	Offered	Group	Language	Activity	Day	Time	Venue	Campus	Study Prog
EBN 111	S1	G01	E	L1 L2 L3	Monday Tuesday Thursday	08:30 - 09:30 13:30 - 14:30 11:30 - 12:30	Roos hall Thuto 3-2 Roos hall	HATFIELD	b2c2e2m2n2p2r2s2z2 B1M1R1
EBN 111	S1	G01	E	T1	Friday	07:30 - 08:30	AE du Toit Auditorium	HATFIELD	b2c2e2m2n2p2r2s2z2 B1M1R1
EBN 111	S1	P01	E	P1	Monday	11:30 - 14:30	Eng II 4-39 CAEC Lab	HATFIELD	n2p2z2
EBN 111	S1	P02	E	P1	Wednesday	14:30 - 17:30	Eng II 4-39 CAEC Lab	HATFIELD	r2s2 B1
EBN 111	S1	P03	E	P1	Thursday	14:30 - 17:30	Eng II 4-39 CAEC Lab	HATFIELD	R1
EBN 111	S1	P04	E	P1	Monday	14:30 - 17:30	Eng II 4-39 CAEC Lab	HATFIELD	b2c2e2m2
EBN 111	S1	P05	E	P1	Friday	09:30 - 12:30	Eng II 4-39 CAEC Lab	HATFIELD	M1
EBN 111	S1	P06	E	P1	Wednesday	11:30 - 14:30	Eng II 4-39 CAEC Lab	HATFIELD	M1

5. STUDY MATERIALS AND PURCHASES

5.1. Prescribed textbooks

Fundamentals of Electric Circuits,
7th Edition

By Charles Alexander and Matthew Sadiku

ISBN10: 1260226409

ISBN13: 9781260226409

5.2. Components

Each student requires their own practical kit (cables, components, and breadboard) for the practicals. The cost of the practical kit is included in the class fees. Arrangements will be made for students to receive the kits. Students who repeat the module do not receive a new kit.

6. LEARNING ACTIVITIES

6.1. Contact time and learning hours

Lectures, tutorials and practicals (contact time)					
	Contact time per week (periods)	Preparation per week (periods)	Total time per week (hours)	Number of weeks	Total time per semester (hours)
Lectures*	3	3	5	12	60
Tutorials*	1	3	3.3	12	40
Practicals	1	1	1.5	12	18
Assignments and assessment (non-contact time)					
	Assessment time (hours)	Preparation time (hours)			Total time per semester (hours)
Assignment(s)	0	8			8
Semester test 1	1.5	7.5			9
Semester test 2	1.5	7.5			9
Examination	3	15			18
Module hours					162
Module credits					16.2

Note 1: The time allocation in this table represents nominal hours, i.e. some students might require less and others more time to attain the pass requirements.

* Note 2: As class tests are written during lecture or tutorial periods, the times allocated for class tests are included in the total lecture or tutorial time.

6.2. Lectures

Refer to the section on lectures in the General EEC Study Guide.

6.2.1 General approach to lectures

Lectures are presented in a style of co-operative and student-centred learning in a face-to-face format **on campus by default**. Students are expected to prepare for classes. Lecture content and structure will be based on the prescribed and recommended study material. Note, however, that various problems and examples that are not in the textbook will be discussed during lectures. Students are expected to prepare for classes as specified in Section 2.1.

6.2.2 Face-to-face lectures

Attendance of lectures is compulsory and a register of attendance will be taken to determine which students bring their part and work consistently. Refer to the general EEC study guide for the departmental strategy to deal with absence from lectures. Important announcements with respect to laboratory sessions, assignments, tutorials and discussions of test content will be made during lecture times. Students need to take note that only selected components of content, assessments and arrangements delivered in face-to-face lectures will be made available online, e.g., announcements regarding tests, assignments, memoranda (optional), lecture notes (optional), etc. Lecturers are not obliged to publish all content delivered during lectures online as a hybrid teaching and learning

strategy will be followed where a significant component of the teaching and learning activities is face-to-face.

6.2.3 Online lectures

As mentioned in Section 2.2, in the case where face-to-face lectures and tutorials are not possible due to specific reasons, pre-recorded videos will be made available via ClickUP. Students are expected to follow the same routine by preparing for a lecture, watching a lecture, and following-up the lecture by completing the necessary textbook problems.

6.3. Hybrid teaching and learning approach

6.3.1 Hybrid teaching and learning

Refer to the section on hybrid teaching and learning in the General EECE Study Guide, and Section 2.3.

6.3.2 Online teaching and learning

The approach delineated in Section 2.2 will be followed should face-to-face contact not be possible.

6.4. Tutorial classes

6.4.1 General approach to tutorial lectures

The *Schedule* with problems to be worked out for each tutorial class will be posted on the module ClickUP page.

Students should solve the prescribed problems *before* the tutorial class. These problems will be discussed during the tutorial class; students should use the class to check if and where they erred in solving problems.

Students who are repeating the module are expected to redo all the tutorials.

6.4.2 Face-to-face tutorials

Please refer to Section 2.

6.4.3 Online tutorials

Please refer to Section 2.

6.5. AMS assignments

A series of online assignments will be released during the semester. The assignments will be released on the Assessment Management System (AMS) (<https://ams.up.ac.za/users/login/>). Details on how to access and complete assignments on the AMS will be provided on ClickUP and in the EECE General Study Guide. Students need to complete the assignments online before their respective deadlines as published on the AMS. Refer to Section 8.4 of the EECE General Study Guide for the online submission protocol related to deadlines. Failure to complete an assignment will result in a mark of zero for the assignment.

The assignments contribute to the semester mark as indicated in Section 7. Failure to submit an assignment will result in a mark of zero for the particular assignment. There will not be any aegrotat assignment opportunities. Students who are repeating the module are expected to redo all the assignments.

If there is suspicion that any part(s) of the work has been copied from another student's work or from another source, the relevant UP policies on plagiarism and academic dishonesty will be followed. Refer to Section 9.2 in the General EECE Study Guide for more information about academic misconduct and the processes that are required to deal with such incidents.

You will be required to sign the following declaration on the front page of your assignments:

I, name and surname, hereby declare that the work herewith is completely my own and that no parts have been copied in any way from current or previous students or other sources.

The use of a chatbot such as ChatGPT is allowed, but needs to be acknowledged explicitly.

6.6.Laboratory work

6.6.1 General approach and practical guide

Approach: Model 1: Laboratory session practicals

As per the General EECE Study Guide, the module will make use of Model 1 for practicals. In other words, the practical time is divided into three 3-hour laboratory sessions during which students conduct experiments and complete a laboratory assignment.

A practical is considered as the combination of the pre-practical assignment and the laboratory assignment. Each laboratory session is preceded by a compulsory pre-practical assignment. As indicated in Section 6.1, three hours of preparation time is necessary to complete a pre-practical assignment.

Practicals:

1. Practical 1: Turning on an LED in a resistor network
2. Practical 2: Using a transistor as a switch to turn on an LED
3. Practical 3: Using an operational amplifier to turn on an LED

The practical guides are made available on ClickUP. The pre-practical and laboratory assignments are completed and evaluated on the AMS. Note that students are encouraged to work in groups of two during laboratory session, but will complete the pre-practical and laboratory assignments on the AMS on an **individual** basis.

Students repeating the module: If a student is repeating the module and has not been granted practical exemption, the student may not use the practical work or reports from the previous year.

Rules and requirements: Each student must do three practicals (pre-practical & laboratory assignment). In order to pass this module,

1. all practicals must be done,
2. an average mark of 50% or more must be obtained overall for the practicals, and
3. a subminimum of 40% must be attained for each practical.

Laboratory groups and sessions: Students will be divided into large laboratory session groups by engineering study discipline (e.g., electronic, mechanical) at the start of the semester. The dates and times each group has to do a particular laboratory session will appear on ClickUP. The responsibility rests on each student to make sure when laboratory session are scheduled for them.

Permission to be moved from a laboratory session group must be obtained in writing from the Assistant Lecturers at least one week before the date of the laboratory session. Failure to pre-arrange a move from one group to another will mean that the student will not be able to do the laboratory session and will subsequently not be eligible for entry into the examination in the module.

A student needs to do all practicals and to be admitted to the exam. In the event of absence from a practical, please refer to the protocol in Section 8.5 of the EECE General Study Guide.

During laboratory sessions: Students must be well prepared when reporting for a laboratory session. The information on ClickUP as well as the relevant material in the textbook must be consulted beforehand. Each student must bring their own textbook to the practical. Students must obey all laboratory rules.

Information regarding practicals: Specific information and about practicals will be posted on ClickUP at least two weeks prior to the commencement of the particular practical. General announcements and reminders will also be made on ClickUP.

Use of chatbots: The use of a chatbot such as ChatGPT is allowed, but needs to be acknowledged explicitly.

Practical fallback strategy for single students

The practical sessions in this module are completed individually. As such a single student fallback strategy is not applicable.

Contingency measure for students unable to attend on-campus practicals owing to serious unforeseen circumstances

Students who are unable to participate in on-campus practicals in the event that such practicals can be presented, must contact the lecturer to discuss a contingency arrangement.

6.6.2 Approach to online practicals

The approach to practicals described above assumes practicals can be completed on campus. If the situation warrants, practicals will be completed online. Any amendments to the submission and completion of practicals will be communicated on ClickUP.

7. RULES OF ASSESSMENT

Refer to the section on assessment in the General EECE Study Guide. Also see the examination regulations in the Year Books of the Faculty of Engineering, Built Environment and Information Technology.

7.1. Requirements, subminima and calculation of marks

Primary Requirements	1. The examination schedule per module is compiled in accordance with the Examination Regulations of the Faculty, the General Examination regulations of the University and the Policy on Examinations and related matters. 2. Admission to examination: To be admitted to the final examination in a module, a semester/year mark of at least 40% is required, with the exception of first year modules on the first semester level, where a semester mark of at least 30% is required. Completion of all assessments that count towards the semester mark is mandatory for entry into the examination of a module. In addition, all
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	other examination admission requirements must be complied with, as applicable to the specific module. 3. Minimum examination mark: In order to pass a module a student must obtain an examination mark of at least 40% and a final mark of at least 50%. 4. ECSA graduate attribute requirements: In modules where ECSA GAs are assessed at exit level, the relevant GA(s) must be passed in addition to the general pass requirements stated above to pass the module. GA pass requirements are stated in the details of the assessment schedule.
Calculation of final mark	Semester mark: 50% Examination mark: 50%
Calculation of semester (year) mark I = individual; G = group	Semester test 1: 35% (I) Semester test 2: 35% (I) Assignments: 6% (G) Class tests: 12% (I) Practicals: 12% (G)
Specific requirements	All scheduled practical assignments must be completed to a satisfactory level. All semester projects and tasks must be completed to a satisfactory level as per Departmental rules. Failure to comply with any of the above will result in examination refusal. Additional conditions: 1. Semester tests 1 and 2 are compulsory. 2. A voluntary third semester test (Semester Test 3) may be written at the end of the semester. The highest two marks of the three written semester tests will be used to calculate the semester mark. No sick test will be awarded for the optional Semester Test 3. Students that have qualified to write a sick test for Semester Tests 1 and/or 2, may be required to write a sick test at the same time as Semester Test 3. As such, students who qualified to write a sick test will not be eligible to write Semester Test 3.
Subminimum requirements	As per primary requirements above. Additional subminimum requirement (semester): 1. An average practical mark of at least 50% must be attained. 2. A subminimum of 40% must be attained for each practical assignment. Based on the qualification criteria in the practical guide, one additional demonstration opportunity will be provided for each practical assignment for students who participated in the practical and did not comply with the 40% practical subminimum for the practical. The average practical mark of students who make use of the additional opportunity will be capped at 50%. Additional subminimum requirements (examination/final mark): None
Requirements for promotion	No promotion allowed.

7.2.Assessment details

Assessment approach

Assessment of class tests, semester tests and the exam will take place electronically via the AMS (Assessment Management System). See module ClickUP page for details.

Practicals

A practical is the combination of a pre-practical assignment and a laboratory assignment. Each assignment is submitted and evaluated via AMS.

Class tests

Three (short) class tests will be written in the semester. The dates for the tests will be announced in class and on the ClickUP page. A mark of zero will be awarded if a student misses a class test, unless the student can provide a valid medical certificate. For legitimate absence the test does not count towards the semester mark. Illegitimate absence from a class test does not result in exam refusal, but the test counts towards the semester mark.

Semester tests

Two compulsory tests of 90 minutes each will be written during the scheduled test weeks of the School of Engineering.

For a legitimate absence from these tests as supported by the necessary documentation (see Section 8 below), an aegrotat test will be written in the final week of lectures.

A voluntary third test of 90 minutes will be written. The mark for this test will replace the lowest mark of the first two tests. There is no aegrotat test for the voluntary third test.

The AMS (an online marking system developed by the EECE department) will be applied to mark the class tests, semester tests and exams. The online marking system is accessible via: <https://ams.up.ac.za/>. Each student will be granted an account to access the online marking system, where students can view their marks, marking log, memos of each question, and make queries for any marking discrepancies. A certain marking query period will be allocated after each class test, semester test, and exam. **Students must please ensure that they visit the AMS online marking system during the granted query period.** After the query period, the mark that appears on the module web page for the relevant test will be considered as final. **No late queries on tests/exams are allowed.**

Study material in tests and exams

Tests and examinations are **CLOSED** book for on-campus evaluations.

8. GENERAL

Refer to the EECE General Study Guide (Part 1 of the study guide) for procedures, policies and rules about absence from formal evaluation opportunities and practical sessions, grievance procedures, academic dishonesty and plagiarism.

MODULE INFORMATION

9. MODULE OBJECTIVES AND LEARNING OUTCOMES

9.1. Content and purpose of the module

The general objective of this module is to develop expertise in solving electric and electronic circuits. The topics covered in the module are Ohm's law, Kirchoff's current and voltage laws, voltage and current division, mesh current and node voltage methods, linearity, Thevenin and Norton equivalent circuits, source transformation, power transfer, energy storage elements in circuits (inductors and capacitors), and operational amplifiers and applications. Although circuits will mostly be solved using direct current (DC) sources, the final part of the module will consider methods to solve circuits using alternating current sources (AC).

9.2. Module Outcomes

The critical learning outcomes for the module are qualified through the ECSA graduate attribute (GA) requirements as discussed in the General EECE Study guide. The following skills are developed at the indicated level and a student must demonstrate competence in each to be able to pass the module.

ECSA GA 1 (Entry Level)

Problem solving

Learning outcome: Demonstrate competence to identify, assess, formulate and solve convergent and divergent engineering problems creatively and innovatively.

At the completion of the module, students should be able to solve any DC or AC circuit containing resistors, capacitors, inductors, sources and operational amplifiers using the circuit analysis tools mastered during the module. Assessments during the module are purely problem-based, e.g. students need to solve for either voltages or currents in circuits.

ECSA GA 2 (Entry Level)

Application of scientific and engineering knowledge

Learning outcome: Demonstrate competence to apply knowledge of mathematics, basic science and engineering sciences from first principles to solve engineering problems.

At the completion of the module, students should know, understand and be able to apply basic circuit analysis tools to both DC and AC circuits. The analysis tools include: Ohm's Law, Kirchoff's Laws, Nodal and Mesh Analysis, Superposition, Source Transformation, and Thevenin and Norton equivalents.

9.3.Cognitive level of assessment

Refer to the section on cognitive level of assessment in the General EECE Study Guide.

	%*
1. Knowledge	10
2. Comprehension	10
3. Application	60
4. Analysis	10
5. Synthesis	
6. Evaluation	
7. Other skills**	10

* Estimate of the % of the total assessment, including all forms of assessment applied in this module, devoted to the various levels of cognitive thinking skills and of UP graduate attributes.

** Assessment of UP graduate attributes:

- Practical skills
- Independent and Team working skills
- Time management
- Conceptualize and synthesize knowledge

10. MODULE STRUCTURE

The detailed structure/schedule for the module is available on the module ClickUP page.

The next topics are covered in the module (the allocated time is only approximate):

<i>Study theme and study units</i>	<i>Mode of instruction</i>	<i>Chapter</i>
1. Basic Concepts • Systems of Units • Charge and Current • Voltage • Power and Energy • Circuit Elements • Applications	Lectures, class discussion, tutor classes and self-study.	1
2. Basic Laws • Ohm's Law • Nodes, Branches, and Loops • Kirchhoff's Laws • Series Resistors and Voltage Division • Parallel Resistors and Current Division • Wye-Delta Transformations • Applications	Lectures, class discussion, tutor classes and self-study.	2
3. Methods of Analysis • Nodal Analysis • Nodal Analysis with Voltage Sources • Mesh Analysis • Mesh Analysis with Current Sources • Nodal and Mesh Analysis by Inspection • Nodal Versus Mesh Analysis • Transistors • Applications	Lectures, class discussion, tutor classes and self-study.	3

<i>Study theme and study units</i>	<i>Mode of instruction</i>	<i>Chapter</i>
4. Circuit Theorems • Linearity Property • Superposition • Source Transformation • Thevenin's Theorem • Norton's Theorem • Derivations of Thevenin's and Norton's Theorem • Maximum Power Transfer • Applications	Lectures, class discussion, tutor classes and self-study.	4
5. Operational Amplifiers • Operational Amplifiers • Ideal Op Amp • Inverting Amplifier • Non-inverting Amplifier • Summing Amplifier • Difference Amplifier • Cascaded Op Amp Circuit • Applications	Lectures, class discussion, tutor classes and self-study.	5
6. Capacitors and Inductors • Capacitors • Series and Parallel Capacitors • Inductors • Series and Parallel Inductors • Applications	Lectures, class discussion, tutor classes and self-study.	6
7. Sinusoids and Phasors • Sinusoids • Phasors • Phasor Relationship for Circuit Elements • Impedance and Admittance • Kirchhoff's Laws in the Frequency Domain • Impedance Combinations • Applications	Lectures, class discussion, tutor classes and self-study.	9
8. Sinusoidal Steady-State Analysis • Nodal Analysis • Mesh Analysis • Superposition Theorem • Source Transformation • Thevenin and Norton Equivalent Circuits • Applications	Lectures, class discussion, tutor classes and self-study.	10
9. Laboratory work	Computer-assisted and experimental work in laboratories.	

11. STUDY THEMES

11.1. STUDY THEME 1: Basic Concepts

11.1.1. Learning outcomes

The student will:

- Understand the basic SI units and SI prefixes
- Learn about the concepts of charge, current, and voltage
- Learn about power and energy
- Become familiar with the passive sign convention
- Learn about the circuit elements including dependent and independent sources

11.1.2. Study units

Chapter 1, section 1.1: Introduction

Chapter 1, section 1.2: SI units

Chapter 1, section 1.3: Charge and Current

Chapter 1, section 1.4: Voltage

Chapter 1, section 1.5: Power and Energy

Chapter 1, section 1.6: Circuit Elements

Chapter 1, section 1.7 and section 1.8: Applications and Problem Solving (self study)

11.1.3. Criteria of assessment

Upon completion of this study theme, the student will:

- **know** the SI units and be able to **determine** derived units.
- know the different equations and units for voltage and current and be able to apply them.
- be able to **define** the basic circuit element.
- **know** the passive sign convention and be able to **apply** it.
- **know** the equation for power and be able to **apply** it.
- be able to **derive** the equation for power in terms of the basic circuit element's terminal voltage and current and be able to **apply** it.
- **know** the characteristics of ideal voltage and current sources.
- be able to **apply** and **interpret** the passive sign convention in power calculations.
- be able to **compute** charge, voltage, current, power and energy for an element.

11.2. STUDY THEME 2: Basic laws

11.2.1. Learning outcomes

The student will:

- Learn about Ohm's law
- Understand the concepts of nodes, branches, and loops
- Learn Kirchhoff's current and voltage laws
- Learn about combining resistors in series and in parallel
- Learn about the principles of voltage division and current division
- Become familiar *with wye-delta transformation*

11.2.2. Study units

Chapter 2, section 2.2: Ohm's Law

Chapter 2, section 2.3: Nodes, Branches, and Loops

Chapter 2, section 2.4: Kirchhoff's Law

Chapter 2, section 2.5: Series Resistors and Voltage Division

Chapter 2, section 2.6: Parallel Resistors and Current Division

Chapter 2, section 2.7: Wye-Delta Transformations

Chapter 2, section 2.8: Applications (self study)

11.2.3. Criteria of assessment

Upon completion of this study theme, the student will:

- **know** the five ideal circuit elements.
- be able to **apply** the definitions of ideal sources and hence **determine** the validity of connected sources.
- **understand** the way dependent sources are handled.
- **understand** the difference between passive and active elements.
- **know** Ohm's law and be able to **apply** it.
- **know** the equations for power in terms of resistance and conductivity and be able to **apply** them.
- be able to **construct** circuit models and **identify** shortcomings with respect to models of practical systems.
- be able to **identify** nodes, loops and meshes.
- be able to **define** direct current, series, parallel and other basic concepts.
- be able to **derive** equations for elements equivalent to elements connected in series and/or parallel.
- be able to **solve** problems involving voltage and current division.
- be able to **design** voltmeters and ammeters.
- **know** Kirchhoff's voltage and current laws and be able to **apply** them.
- be able to **derive** equations for delta-to-Y conversions and **apply** them

11.3. STUDY THEME 3: Methods of Analysis

11.3.1. Learning outcomes

The student will:

- Learn about nodal and mesh analysis
- Understand the concept of supernode and supermesh
- Apply what is learnt to DC transistor circuits

11.3.2. Study units

Chapter 3, section 3.2: Nodal Analysis

Chapter 3, section 3.3: Nodal Analysis with Voltage Sources

Chapter 3, section 3.4: Mesh Analysis

Chapter 3, section 3.5: Mesh Analysis with Current Sources

Chapter 3, section 3.6: Nodal and Mesh Analysis by Inspection

Chapter 3, section 3.7: Nodal Versus Mesh Analysis

Chapter 3, section 3.9: Applications: DC Transistor Circuits (self study)

11.3.3. Criteria of assessment

Upon completion of this study theme, the student will be able to:

- determine a set of independent simultaneous equations consisting of node-voltage equations and constraint equations from which a circuit can be solved.
- **identify** a supernode and **formulate** a new set of equations associated with the supernode.

- **determine a set of** independent simultaneous equations consisting of mesh-current equations and constraint equations from which a circuit can be solved.
- **identify** a supermesh and **formulate** a new set of equations associated with the supermesh.
- **identify** the constraint equations and **formulate** the new set of equations associated with dependant sources for the mesh-current method.

11.4. STUDY THEME 4: Circuit Theorems

11.4.1. Learning outcomes

The student will:

- Learn about theorems used for simplifying analysis
- Understand the principles of linearity and superposition
- Learn of about the concepts of source transformation and equivalence.
- Be able to use Thevenin's theorem to find a Thevenin equivalent circuit.
- Be able to use Norton's theorem to find a Norton equivalent circuit.
- Understand the concept of maximum power transfer to the load.

11.4.2. Study units

Chapter 4, section 4.2: Linearity Property

Chapter 4, section 4.3: Superposition

Chapter 4, section 4.4: Source Transformation

Chapter 4, section 4.5: Thevenin's Theorem

Chapter 4, section 4.6: Norton's Theorem

Chapter 4, section 4.7: Derivations of Thevenin's and Norton's Theorem

Chapter 4, section 4.8: Maximum Power Transfer

Chapter 4, section 4.10: Applications (self study)

11.4.3. Criteria of assessment

Upon completion of this study theme, the student will be able to:

- **use** the linearity property to **calculate** circuit variables
- **use** the superposition principle to **calculate** circuit variables
- **derive** a source transformation for a given circuit and **use** source transformation to **calculate** circuit variables.
- **calculate** Thévenin and Norton equivalent circuits by **determining** which of the following methods is applicable and then **apply** it:
 - **determine** V_{Th} and R_{Th} by **calculating** the open-circuit voltage and short-circuit current.
 - **apply** source transformations
 - **determine** R_{Th} by the method of deactivating independent sources
 - **calculate** R_{Th} by deactivating independent sources and adding a test source for circuits containing dependant sources only.
- **use** the superposition principle to **derive** Thevenin's and Norton's Theorems
- **derive** the equations associated with maximum power transfer and **apply** the underlying principles to other circuits.
- **apply** the method of superposition to linear circuits.

11.5. STUDY THEME 5: Operational amplifiers.

11.5.1. Learning outcomes

The student will:

- Understand what operational amplifiers are
- Learn about the ideal op amp
- Learn some op amp circuits including inverting, noninverting, summing, and difference amplifiers

11.5.2. Study units

Chapter 5, section 5.2 and section 5.3: Operational Amplifier and Ideal Op Amp

Chapter 5, section 5.4: Inverting Amplifier

Chapter 5, section 5.5: Non-Inverting Amplifier

Chapter 5, section 5.6: Summing Amplifier

Chapter 5, section 5.7: Difference Amplifier

Chapter 5, section 5.8: Cascaded Op Amp Circuit

Chapter 5, section 5.10: Applications (self study)

11.5.3. Criteria of assessment

Upon completion of this study theme, the student will be able to

- **derive** transfer characteristics of basic op-amp circuits (e.g., inverting-amplifier circuit, summing-amplifier circuit, noninverting-amplifier circuit, and difference-amplifier circuit) from first principles.
- completely **solve** circuits in which no more than four op-amps are connected.

11.6. STUDY THEME 6: Capacitors and Inductors

11.6.1. Learning outcomes

The student will:

- Learn about capacitors and capacitance
- Understand how to combine capacitors in series and parallel
- Learn about inductors and inductance
- Understand how to combine inductors in series and parallel

11.6.2. Study units

Chapter 6, section 6.2: Capacitors

Chapter 6, section 6.3: Series and Parallel Capacitors

Chapter 6, section 6.4: Inductors

Chapter 6, section 6.5: Series and Parallel Inductors

Chapter 5, section 6.6: Applications (self study)

11.6.3. Criteria of assessment

Upon completion of this study theme, the student will be able to

- **solve** time-varying voltages and currents in circuits containing combinations of either inductors or capacitors.
- **Compute** energy and power associated with the inductors and capacitors in the above problems.

11.7. STUDY THEME 7: Sinusoids and Phasors

11.7.1. Learning outcomes

The student will:

- Understand the concepts of sinusoids and phasors

- Apply phasors to circuit elements
- Be introduced to the concepts of impedance and admittance
- Learn about impedance combinations

11.7.2. Study units

Chapter 9, section 9.2: Sinusoids

Chapter 9, section 9.3: Phasors

Chapter 9, section 9.4: Phasor Relationship to Circuit Elements

Chapter 9, section 9.5: Impedance and Admittance

Chapter 9, section 9.6: Kirchhoff's Laws in the Frequency Domain

Chapter 9, section 9.7: Impedance Combinations

Chapter 9, section 9.8: Applications (self study)

11.7.3. Criteria of assessment

Upon completion of this study theme, the student will be able to:

- **compute** the phasor representation of sinusoidal currents and voltages and also determine inverse phasor transforms.
- **compute** the rms-value of a periodic function.
- **Apply** Kirchhoff's Law in the frequency domain

11.8. STUDY THEME 8: Sinusoidal Steady-State Analysis

11.8.1. Learning outcomes

The student will:

- Apply previously learnt circuit techniques to sinusoidal steady-state analysis
- Learn how to apply nodal and mesh analysis in the frequency domain
- Learn how to apply superposition, Thevenin's and Norton's theorems in the frequency domain

11.8.2. Study units

Chapter 10, section 10.2: Nodal Analysis

Chapter 10, section 10.3: Mesh Analysis

Chapter 10, section 10.4: Superposition Theorem

Chapter 10, section 10.5: Source Transformation

Chapter 10, section 10.6: Thevenin and Norton Equivalent Circuits

Chapter 10, section 10.7: Applications (self study)

11.8.3. Criteria of assessment

Upon completion of this study theme, the student will be able to:

- **compute** the frequency domain equivalent circuit of any circuit containing sinusoidal sources and combinations of resistors, inductors, and capacitors.
- **solve** frequency domain circuits by using standard circuit techniques.

11.9. STUDY THEME 9: Practicals

11.9.1. Learning outcomes

The student will:

- Analyse a set of circuits using previously learnt circuit theorems

- Simulate the circuits with an appropriate computer assisted design tool.
- Build the circuits on an experimental board and evaluate the circuits in a laboratory setup.

11.9.2. Criteria of assessment

Upon completion of this study theme, the student will be able to:

- **use** computer aided software tools to analyse a circuit.
- **use** basic laboratory tools to evaluate a physical circuit.

12. PRACTICALS

Refer to Section 6.6 in the organisational section of this guide for a description of the approach to the practical work in this module. Details regarding practicals will be communicated on the module ClickUP page.